DATA ANALYTICS / BUSINESS INTELLIGENCE

NYC Airbnb Market Analysis

2019 Listing Supply, Pricing, Room-Type Mix & Host Engagement across 48,895 New York City listings

PREPARED BY MD OSAMA | PYTHON • PANDAS • TABLEAU

01 / EXECUTIVE SUMMARY

Concentrated Market with Clear Pricing Tiers

Manhattan and Brooklyn lead listing supply while Manhattan commands top nightly asking rates.

48,895

Total Listings

Active 2019 records

37,457

Distinct Hosts

Unique host IDs

$152.72

Avg. Nightly Price

Citywide median: $106

1.14M

Total Reviews

1,138,005 cumulative

112.8d

Avg. Availability

Calendar days open

Strategic Signals & Interpretations

85.4% Core Supply

Prioritize Manhattan and Brooklyn benchmarks.

2.25x Price Spread

Compare local room segments over averages.

Descriptive Proxies

Reviews do not equal realized revenue.

Market Questions & Analytical Workflow

BUSINESS QUESTION	ANALYTICAL MEASURE / CUT
Where is supply concentrated?	Listing count and percentage share by borough
How do asking prices vary?	Average and median price by borough and room type
Which listings show engagement?	Cumulative review counts across high-volume hosts
How does availability differ?	Reported calendar availability days by segment

STEP 01

Inspect

48,895 records across 16 fields

STEP 02

Clean

Audit missingness and edge cases

STEP 03

Explore

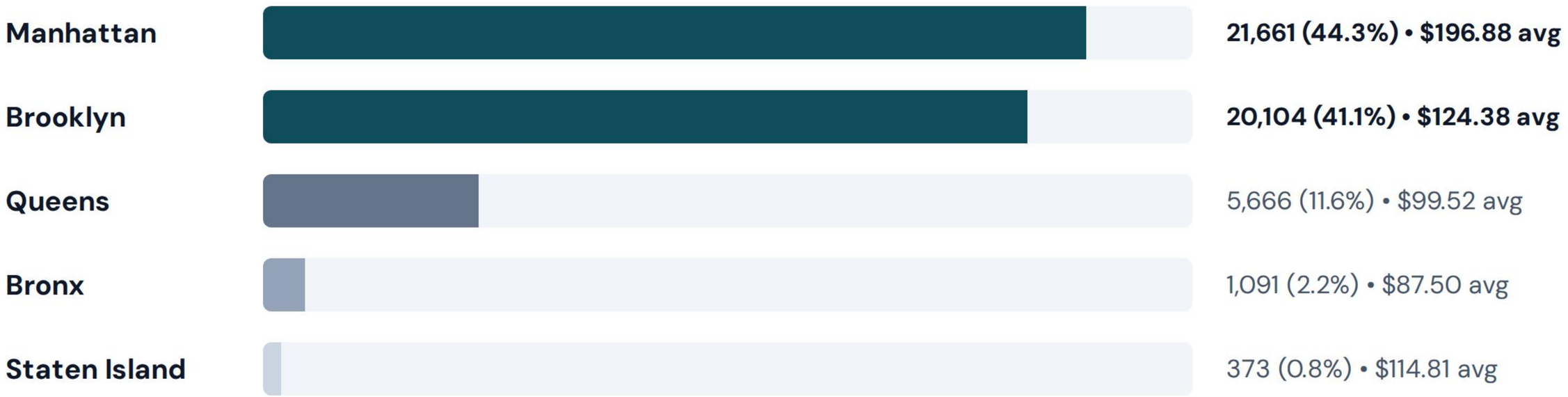
Segment borough, room, price

STEP 04

Communicate

Synthesize findings in Tableau

Manhattan & Brooklyn Drive 85.4% of Supply



Takeaway: Market entry decisions require neighborhood-level competition and room-type mix analysis.

04 / PRICING DYNAMICS

Manhattan Commands Highest Nightly Rates

BOROUGH	AVG NIGHTLY PRICE	MEDIAN PRICE	AVG AVAILABILITY
Manhattan	**$196.88**	$150.00	112.0 days
Brooklyn	$124.38	$90.00	100.2 days
Staten Island	$114.81	$75.00	199.7 days
Queens	$99.52	$75.00	144.5 days
Bronx	$87.50	$65.00	165.8 days

MANHATTAN VS BRONX

2.25x Spread

OVERALL MEDIAN

$106 / Night

Pair mean prices with medians to control for luxury outliers.

Entire Homes & Private Rooms Equal 97.6%

ROOM TYPE	LISTINGS	SHARE	AVG PRICE	MEDIAN
Entire home / apt	25,409	**52.0%**	$211.79	$160
Private room	22,326	**45.7%**	$89.78	$70
Shared room	1,160	2.4%	$70.13	$45

$211.79

Entire Home Premium

Benchmark pricing strictly within identical room categories.

Signals vs. Realized Performance Outcomes

Supported by Dataset

Review Volume: Compare cumulative counts across hosts.

Top Performers: Flag high-review listings for study.

Calendar Supply: Contrast open days across segments.

Requires External Data

True Occupancy: Requires verified booked-night logs.

Net Revenue: Needs realized rates and operating costs.

Guest Satisfaction: Demands star ratings and sentiment.

Data Quality Note: 10,052 rows (20.6%) lack last_review and reviews_per_month values.

Decision Agenda for Next Analysis Cycle

01. Benchmark Neighborhoods: Compare median prices by local district and room type.

02. Segment Product Tiers: Build comparable-listing price bands and isolate outliers.

03. Audit Engagement: Incorporate ratings, review recency, and text sentiment.

04. Measure Realized Yield: Integrate booked nights, transaction rates, and host costs.

05. Operationalize KPIs: Track supply shares, price medians, and review velocity.

Baseline Capabilities & Analytical Guardrails

Core Supply

Concentrated in Manhattan and Brooklyn.

Price Leader

Manhattan leads average asking prices.

Historical Scope

2019 snapshot excludes current dynamics.

Proxy Limits

Asking rates differ from realized bookings.

FINAL TAKEAWAY

Segment by Borough & Room Type Before Drawing Revenue Conclusions

Validate asking prices against comparable local listings and integrate booking and cost data to measure true profitability.

MD OSAMA • DATA ANALYSIS & BUSINESS INTELLIGENCE • SOURCE: AB_NYC_2019.CSV